Narin Dhatwalia

Sr. Technical Architect, AI/ML
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Summary

Technical Architect with 6+ years building and deploying enterprise AI/ML systems for Fortune 500 clients, including marketing propensity models and Bayesian regression marketing mix models for allocating Kellogg's LATAM spend across Digital, TV, and Print. At Snowflake, take customers from discovery to production on LLM applications, agents, and predictive models, and won first prize at an internal hackathon for a fine-tuning approach now being productized. MS in Machine Learning and Data Science from Northwestern; pursuing a DBA on enterprise GenAI's financial impact.

Work Experience

Snowflake | ***Sr. Technical Architect, AI/ML*** **Jun 2024 - Present**

- Won first prize at an internal Snowflake hackathon by fully fine-tuning an open-weight model (Qwen3-8B) on GPUs using responses from a frontier-lab teacher LLM. Compared the two on an unseen test set for latency, cost, and quality: lower cost and latency with comparable quality. Leadership and Product & Engineering have backed it as a native fine-tuning experience for customers, and I was selected to present to the CEO.
- Architected a production Cortex AI pipeline leveraging Claude 3.5 Sonnet to analyze clinical data for ~400,000 patients and recommend HCC billing codes with supporting clinical reasoning, projecting ~$34M in new revenue and supporting a $7.5M contract renewal. Built an LLM evaluation framework for accuracy, groundedness, and context relevance to monitor outputs.
- Designed a Snowflake CoWork solution for a financial services client, replacing manual SQL workflows with a unified Cortex Agent that combines text-to-SQL over a bespoke semantic layer with search over unstructured CRM notes, returning answers in seconds. Instrumented it with evals and AI observability (cost, latency, user feedback, prompt and response traces), enforced governance with Row Access Policies, and worked with Snowflake Product & Engineering on private preview features. Defined the DEV-to-PROD promotion gate for agents with an LLM-as-judge eval suite and pass/fail criteria, replacing subjective spot checks.
- Built ML models across industries: XGBoost router outage prediction with time-series cross-validation for a telecommunications client (40% less network downtime; POC contributed to a $7.2M Snowflake deal), new-release revenue prediction on a streaming platform for a media client, and transaction categorization for a fintech client covering ~50M payment records weekly.
- Built a text-to-SQL chatbot with Cortex Analyst for a major SaaS company so Customer Success Managers can query account health data in natural language, reducing SQL bottlenecks by 80%.
- Lead technical workshops, executive readouts, and advisory for Fortune 500 customers; support pre-sales as needed (discovery, solution design, estimation, SOWs). Spoke at an internal summit (150+ attendees); authored technical content used by customers.

TransUnion | ***Consultant, Data Science & Analytics*** **Jan 2023 - Jun 2024**

- Built SQL- and Spark-based attribute sets measuring consumer creditworthiness for originations-risk, account management, and marketing propensity models; owned "Soft Inquiry Attributes" end to end, including R-based feature calculation and performance-lift analysis for credit card marketing.
- Led a 6-month Data Science Innovation study comparing GPU-accelerated Random Forest against the XGBoost-based originations baseline: significantly lower training time with less overfitting.
- Developed an originations-risk underwriting model using Survival Analysis that predicts the month of potential delinquency, raising AUC from 0.80 to 0.84 and outperforming legacy models.

Decision Point Analytics | ***Data Scientist*** **Feb 2020 - Jun 2021**

- Built Bayesian regression Market Mix Models for Kellogg's LATAM to allocate marketing spend across Digital, TV, and Print, incorporating expert input through conjugate priors.
- Used Facebook's Prophet to optimize Coca-Cola Mexico's promotions, decomposing volume lift into forward buying, cannibalization, and true lift, driving a 20% revenue boost the following quarter.

Antuit.ai | ***Business Analyst*** **Dec 2017 - Sep 2018**

- Built automated ETL pipelines across Nielsen, NPD, and IRi data plus Tableau/Power BI dashboards, cutting turnaround by 6 days.

Projects

Text-to-SQL CLI Agent | GitHub portfolio

- Built a conversational text-to-SQL agent for SQLite databases using an open-weight model served by Fireworks AI; added schema grounding, execution-guided SQL self-repair, and a two-layer guardrail. Evaluated outputs by executing generated queries and comparing result rows, achieving 9/10 accuracy on the evaluation set with 1.15s median latency versus 1/10 for a naive prompt on the same model.

Generative AI: Infinite Shakespeare with Transformers | GitHub portfolio

- Built and trained a decoder-only Transformer language model in PyTorch on Shakespeare's complete works, generating an endless stream of text that closely resembled his plays. Implemented multi-head self-attention, feed-forward blocks, skip connections, layer normalization, and triangular masking for token-level affinities.

Education

Ottawa University | ***Doctorate, Business Administration*** **Dec 2025 - Present**

Northwestern University | ***MS, Machine Learning and Data Science*** **(GPA: 3.926)** **Sep 2021 - Dec 2022**

University of Delhi | ***BS (Honors), Mathematics*** **(GPA: 81.46%)** **Jul 2014 - Jul 2017**

Skills

- **Languages and Data**: SQL (Snowflake, PostgreSQL), Python, R, Spark (distributed computing), pipeline orchestration
- **ML and AI**: Scikit-learn, XGBoost, PyTorch, TensorFlow, LangChain, LangGraph, RAG, agentic architectures (tool integrations, guardrails), MCP integrations, LLM evaluation (LLM-as-judge), fine-tuning and distillation, open-weight serving (Fireworks AI), Machine Learning
- **Snowflake (SnowPro Core Certified)**: Cortex AI (Agents, Search, Analyst), Snowflake CoWork, Snowpark, UDFs/UDTFs, Streamlit, Notebooks, Model Registry, Feature Store, Dynamic Tables, RBAC, Stored Procedures
- **BI and Databases**: Tableau, Power BI, Sigma, Neo4j (Graph Database)
- **AWS**: S3, RDS, ECS, ECR, SageMaker, EC2, VPC, CloudFormation, Route53, AWS CLI
- **Marketing and Growth**: Marketing Mix Modeling, Holdout Testing, Marketing and Growth Data Science, A/B Testing